GasGrid Project Viva Questions & Answers

1. Project Introduction and Basic Questions

Q1. What is GasGrid?

Ans. GasGrid is a Smart LPG Booking and Distribution Management System designed to manage customer registration, LPG cylinder booking, booking history, and distribution-related operations through a web application.

Q2. What is the main objective of your project?

Ans. The main objective is to simplify LPG booking and distribution management and demonstrate the practical use of DBMS and Operating System concepts.

Q3. Why did you choose this project?

Ans. We chose this project because LPG booking is a common real-world activity. It provides a practical way to understand web development, database management, and OS concepts together.

Q4. What are the main modules of GasGrid?

Ans.

- User registration and login
- Customer dashboard
- LPG cylinder booking
- Booking history and delivery status
- Database management
- OS simulations using C

Q5. What is the problem your project solves?

Ans. It provides a centralized way to manage customer information and LPG bookings, reducing manual record-keeping and making information easier to access.

Q6. Who are the users of your system?

Ans. Customers can book cylinders and view booking details, while administrators can manage bookings and related records, depending on the implemented access permissions.

Q7. What is the scope of your project?

Ans. It is an academic prototype that demonstrates LPG booking operations, database management, and selected OS concepts. It is not necessarily connected to a real LPG distributor.

Q8. What makes your project different from a simple website?

Ans. It combines a web interface and backend logic with database operations and separate C programs demonstrating Operating System concepts.

2. Technology Stack Questions

Q9. Which technologies are used in GasGrid?

Ans. HTML, CSS, and JavaScript for the frontend, Java for the backend, Oracle Database 11g XE with SQL and PL/SQL for the database, and C for OS simulations.

Q10. Why did you use HTML?

Ans. HTML provides the structure of web pages, such as forms, buttons, headings, tables, and input fields.

Q11. What is the role of CSS?

Ans. CSS controls the appearance of the website, including colors, fonts, spacing, layout, and responsiveness.

Q12. Why is JavaScript used?

Ans. JavaScript adds interactivity to the frontend, such as form validation, button actions, and dynamic updates.

Q13. What is the role of Java in your project?

Ans. Java handles backend operations, processes application requests, applies business logic, and can communicate with the database through JDBC.

Q14. Why did you choose Java?

Ans. Java is platform-independent when a compatible JVM is available, supports object-oriented programming, and provides libraries for database connectivity.

Q15. Why did you use C for Operating System concepts?

Ans. C provides low-level programming features and is commonly used to demonstrate process scheduling, synchronization, and resource management concepts.

Q16. What is the difference between frontend and backend?

Ans. The frontend is the part users interact with. The backend processes requests, applies business rules, and handles data operations.

Q17. What is an API?

Ans. An API is an interface that allows two software components to communicate and exchange data.

Q18. What is JDBC?

Ans. JDBC stands for Java Database Connectivity. It allows Java programs to connect to databases, execute SQL statements, and retrieve results.

Q19. What is the JVM?

Ans. JVM stands for Java Virtual Machine. It executes Java bytecode.

Q20. What is the difference between Java and JavaScript?

Ans. Java is a general-purpose programming language commonly used for backend applications. JavaScript is widely used to add interactivity to websites. They are different languages despite their similar names.

3. DBMS and Oracle Questions

Q21. What is DBMS?

Ans. DBMS stands for Database Management System. It is software used to store, organize, retrieve, and manage data.

Q22. What is RDBMS?

Ans. RDBMS stands for Relational Database Management System. It stores data in related tables consisting of rows and columns.

Q23. Why did you use Oracle Database 11g XE?

Ans. Oracle Database 11g XE is a database platform suitable for academic projects. It supports SQL, PL/SQL, constraints, transactions, and relational data management.

Q24. What is SQL?

Ans. SQL stands for Structured Query Language. It is used to create tables, insert records, retrieve data, update records, and delete records.

Q25. What is PL/SQL?

Ans. PL/SQL is Oracle's procedural extension of SQL. It supports variables, conditions, loops, procedures, functions, and exception handling.

Q26. What is the difference between SQL and PL/SQL?

Ans. SQL is mainly used to perform database operations. PL/SQL combines SQL statements with procedural programming features.

Q27. What is a table?

Ans. A table stores related data in rows and columns. For example, a CUSTOMERS table can store customer IDs, names, and contact details.

Q28. What is a primary key?

Ans. A primary key uniquely identifies each row in a table. It cannot contain duplicate or NULL values.

Q29. What is a foreign key?

Ans. A foreign key establishes a relationship between tables by referencing a primary key or another eligible unique key.

Q30. What is the difference between a primary key and a foreign key?

Ans. A primary key uniquely identifies a record in its own table. A foreign key links a record to a related record in another table or the same table.

Q31. What is normalization?

Ans. Normalization organizes database tables to reduce unnecessary data duplication and prevent insertion, update, and deletion anomalies.

Q32. What is a constraint?

Ans. A constraint is a rule applied to a table or column to maintain data integrity. Examples include PRIMARY KEY, FOREIGN KEY, NOT NULL, UNIQUE, and CHECK.

Q33. What are DDL, DML, DQL, DCL, and TCL?

Ans.

Type	Purpose	Examples
DDL	Defines database structure	CREATE, ALTER, DROP
DML	Modifies records	INSERT, UPDATE, DELETE
DQL	Retrieves records	SELECT
DCL	Controls permissions	GRANT, REVOKE
TCL	Manages transactions	COMMIT, ROLLBACK, SAVEPOINT

Q34. What is a JOIN?

Ans. A JOIN combines related rows from two or more tables based on a specified condition.

Q35. What is the difference between DELETE, DROP, and TRUNCATE?

Ans.

- DELETE removes selected or all rows.
- DROP removes the table structure and its data.
- TRUNCATE removes all rows while retaining the table structure.

Q36. What is a transaction?

Ans. A transaction is a logical unit of database work. It may include multiple operations that should be handled consistently.

Q37. What are ACID properties?

Ans.

- Atomicity: All operations succeed or the transaction is rolled back.
- Consistency: The database remains valid.
- Isolation: Concurrent transactions are controlled to prevent improper interference.
- Durability: Committed changes persist.

Q38. What is a database sequence in Oracle?

Ans. A sequence generates numeric values, often used to assign unique IDs to records.

Q39. What is a stored procedure?

Ans. A stored procedure is a named PL/SQL program stored in the database that can perform a specific operation.

Q40. What is exception handling in PL/SQL?

Ans. It handles runtime errors using an EXCEPTION block, allowing the program to respond to errors rather than terminating unexpectedly.

4. SQL Queries You Should Know

These are common practical questions a teacher may ask you to write on the board.

Q41. How do you create a customer table?

```
CREATE TABLE CUSTOMERS (
   CUSTOMER_ID NUMBER PRIMARY KEY,
   NAME VARCHAR2(50),
   PHONE VARCHAR2(15)
);
```

Q42. How do you insert a customer record?

```
INSERT INTO CUSTOMERS
VALUES (1, 'Kritika', '9876543210');
```

Q43. How do you display all customers?

```
SELECT * FROM CUSTOMERS;
```

Q44. How do you update a customer's phone number?

```
UPDATE CUSTOMERS
SET PHONE = '9123456780'
WHERE CUSTOMER_ID = 1;
```

Q45. How do you delete a booking?

```
DELETE FROM BOOKINGS
WHERE BOOKING_ID = 1;
```

Q46. How do you count total bookings?

```
SELECT COUNT(*) FROM BOOKINGS;
```

Q47. How do you display bookings for a particular customer?

```
SELECT *
FROM BOOKINGS
WHERE CUSTOMER_ID = 1;
```

Q48. How do you sort bookings by booking ID?

```
SELECT *
FROM BOOKINGS
ORDER BY BOOKING_ID;
```

Q49. How do you join customers and bookings?

```
SELECT C.NAME, B.BOOKING_ID
FROM CUSTOMERS C
JOIN BOOKINGS B
ON C.CUSTOMER_ID = B.CUSTOMER_ID;
```

Note: These are illustrative queries. Your actual table and column names may differ.

5. Operating System Questions

Q50. What is an Operating System?

Ans. An Operating System is system software that manages computer hardware, memory, processes, files, and other resources.

Q51. Why are OS concepts included in GasGrid?

Ans. The C module demonstrates how Operating System concepts such as scheduling, synchronization, and resource allocation can be represented using practical examples.

Q52. What is a process?

Ans. A process is a program in execution. It has its own execution state and allocated resources.

Q53. What is a thread?

Ans. A thread is a unit of execution within a process. Threads belonging to the same process can share its resources.

Q54. What is CPU scheduling?

Ans. CPU scheduling is the process of selecting which ready process should execute on the CPU next.

Q55. Name common CPU scheduling algorithms.

Ans.

- FCFS (First Come, First Served)
- SJF (Shortest Job First)
- Priority Scheduling
- Round Robin

Q56. What is FCFS?

Ans. FCFS executes processes in the order they arrive. It is simple but may cause long waiting times.

Q57. What is Round Robin scheduling?

Ans. Round Robin assigns each process a fixed time quantum. If a process does not finish within that time, it returns to the ready queue.

Q58. What is a critical section?

Ans. A critical section is a part of a program that accesses shared data or resources and must be protected against unsafe concurrent access.

Q59. What is a race condition?

Ans. A race condition occurs when the result depends on the timing or order of concurrent operations accessing shared data.

Q60. What is a mutex?

Ans. A mutex is a synchronization mechanism that allows only one thread at a time to hold a particular lock.

Q61. What is a semaphore?

Ans. A semaphore is a synchronization mechanism that uses a counter to control access to shared resources.

Q62. What is the difference between a mutex and a semaphore?

Ans. A mutex generally provides exclusive ownership of a resource. A semaphore can allow a specified number of threads or processes to access a resource.

Q63. What is deadlock?

Ans. Deadlock occurs when processes wait indefinitely for resources held by one another, preventing them from proceeding.

Q64. What are the four necessary conditions for deadlock?

Ans.

1. Mutual exclusion
2. Hold and wait
3. No preemption
4. Circular wait

Q65. What is starvation?

Ans. Starvation occurs when a process waits indefinitely or for an excessively long time because other processes keep receiving the required resources.

Q66. What is the difference between deadlock and starvation?

Ans. In deadlock, a group of processes cannot proceed because of circular resource dependencies. In starvation, a process is repeatedly denied the resources or CPU time it needs.

Q67. What is process synchronization?

Ans. Process synchronization coordinates concurrent processes or threads to ensure safe access to shared resources.

Q68. What is a system call?

Ans. A system call is a mechanism through which a program requests a service from the Operating System, such as creating a process or accessing a file.

Q69. What is memory management?

Ans. Memory management is the OS function responsible for allocating, tracking, and freeing memory for programs.

Q70. What is the difference between a program and a process?

Ans. A program is a set of instructions stored in a file. A process is an instance of a program that is currently executing.

6. C Programming Questions for the OS Module

Q71. Why is C suitable for OS simulations?

Ans. C provides efficient execution, pointers, and direct control over data structures, making it useful for demonstrating OS algorithms.

Q72. What is an algorithm?

Ans. An algorithm is a finite sequence of steps used to solve a problem.

Q73. What is a scheduling time quantum?

Ans. It is the maximum CPU time allocated to a process during one turn in Round Robin scheduling.

Q74. What is a ready queue?

Ans. A ready queue contains processes waiting for CPU execution.

Q75. What is waiting time?

Ans. Waiting time is the total time a process spends waiting in the ready queue.

Q76. What is turnaround time?

Ans. Turnaround time is the total time from a process's arrival until its completion.

$$\text{Turnaround Time}=\text{Completion Time}-\text{Arrival Time}$$

Q77. What is response time?

Ans. Response time is the time between a process's arrival and its first CPU execution.

Q78. What is a header file in C?

Ans. A header file contains declarations of functions, macros, and data types. For example, stdio.h provides standard input/output functions such as printf().

Q79. What is the purpose of if-else in C?

Ans. It allows a program to make decisions based on conditions.

Q80. What is the difference between a simulation and a real OS mechanism?

Ans. A simulation demonstrates the logic of a concept. A real OS mechanism interacts with actual processes, threads, locks, or system resources.

Important for GasGrid: If your C code only changes a variable such as lock_available, it demonstrates the idea of a lock but does not implement a real multithreaded mutex. Be clear about this distinction in the viva.

7. Project Integration and Application Questions

Q81. How does the frontend communicate with the backend?

Ans. The frontend sends user input or requests to the backend through the communication mechanism implemented in the application, such as HTTP requests to an API.

Q82. How does Java communicate with Oracle Database?

Ans. Java can use JDBC with an appropriate Oracle JDBC driver, database URL, username, and password to execute SQL queries.

Q83. What is CRUD?

Ans. CRUD stands for Create, Read, Update, and Delete. These are the four basic operations used to manage records.

Q84. How would you prevent duplicate bookings?

Ans. I would validate booking requests in the backend and use database constraints or transactions where appropriate to prevent duplicate or conflicting records.

Q85. How would you prevent unauthorized access?

Ans. By implementing authentication, role-based authorization, secure password storage, and server-side permission checks.

Q86. What is input validation?

Ans. Input validation checks whether user-provided data meets expected rules before the application processes or stores it.

Q87. What is the difference between authentication and authorization?

Ans. Authentication verifies who the user is. Authorization determines what the authenticated user is allowed to do.

Q88. What is SQL injection?

Ans. SQL injection is an attack in which malicious SQL input alters a database query. Prepared statements and input validation help prevent it.

Q89. What is JDBC connection management?

Ans. It involves establishing database connections, executing queries, handling errors, and closing resources properly.

Q90. What happens if the database connection fails?

Ans. The application should handle the exception, display an appropriate error message, and avoid treating the failed operation as successful.

Q91. Why should database transactions be used for important operations?

Ans. Transactions help ensure that related database changes are completed consistently or rolled back if an operation fails.

Q92. How can OS synchronization relate to LPG booking?

Ans. If multiple users attempt to book the last available cylinder simultaneously, synchronization and transaction control can help prevent inconsistent inventory updates.

Q93. What is concurrency?

Ans. Concurrency means multiple tasks make progress during overlapping periods, potentially sharing resources.

Q94. What is the difference between concurrency and parallelism?

Ans. Concurrency involves managing multiple tasks during overlapping periods. Parallelism means multiple tasks execute at the same time, typically on different processing units.

Q95. What testing should be performed on GasGrid?

Ans. Unit testing, input validation testing, database testing, integration testing, and system testing should be performed.

8. Git, GitHub, and Project Management

Q96. What is Git?

Ans. Git is a distributed version control system used to track code changes.

Q97. What is GitHub?

Ans. GitHub is an online platform for hosting Git repositories and collaborating on software projects.

Q98. What is a commit?

Ans. A commit is a recorded snapshot of changes in a Git repository.

Q99. What is a branch?

Ans. A branch is an independent line of development that allows changes to be developed separately.

Q100. What is the difference between Git and GitHub?

Ans. Git tracks code changes locally and across repositories. GitHub hosts repositories online and provides collaboration tools.

Q101. What is a README file?

Ans. A README file explains the project's purpose, technologies, setup steps, features, and usage instructions.

Q102. Why is version control important for a team project?

Ans. It tracks changes, supports collaboration, helps identify bugs introduced by recent changes, and allows earlier versions to be restored.

9. Tricky Questions Your Teacher May Ask

Q103. Is GasGrid a fully deployed commercial LPG system?

Ans. No. It is an academic project prototype intended to demonstrate booking management and technical concepts. Commercial deployment would require additional security, reliability, and integration work.

Q104. Are your OS simulations the same as the actual Operating System?

Ans. No. They demonstrate selected OS concepts through C programs and do not replace the actual OS scheduler or synchronization mechanisms.

Q105. Why are you using both Java and C?

Ans. Java is used for application-level backend functionality, while C is used separately to demonstrate Operating System concepts as required by the project.

Q106. Why use both SQL and PL/SQL?

Ans. SQL performs database operations, while PL/SQL allows those operations to be combined with procedural logic, conditions, loops, and exception handling.

Q107. What are the limitations of your project?

Ans. As an academic prototype, it may lack real distributor integration, live delivery tracking, production-grade security, and large-scale deployment capabilities.

Q108. What future improvements can you make?

Ans. The system could include payment integration, SMS notifications, real-time delivery tracking, improved inventory management, and more comprehensive security and testing.

Q109. What was the most important learning outcome?

Ans. The project helps connect theoretical DBMS and OS concepts with a practical application, while also providing experience with frontend development, backend logic, and database integration.

Q110. How would you explain the complete project in 30 seconds?

Ans. “GasGrid is a web-based LPG Booking and Distribution Management System developed as an academic project. It uses HTML, CSS, and JavaScript for the frontend, Java for backend operations, Oracle Database 11g XE for data management, and C programs to demonstrate Operating System concepts. The project aims to simplify LPG booking operations while demonstrating the practical application of DBMS and OS concepts.”